

Data Preprocessing and Exploratory Data Analysis (EDA)

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1 Introduction

Natural language text scraped from the web is inherently noisy. Over 50% of the raw IMDB movie reviews contained HTML artifacts (e.g., `<br />`), non-standard punctuation, and inconsistent whitespace. A robust, deterministic preprocessing pipeline was essential to prepare the data for downstream machine learning tasks, ranging from Multinomial Naive Bayes to BERT.

2 Data Extraction

The raw dataset was provided as individual text files nested within subdirectories based on sentiment and split.

- **Binary Classification Data:** Extracted from `train/pos`, `train/neg`, `test/pos`, and `test/neg`. The parent folder dictated the 1 or 0 binary label.
- **Continuous Regression Data:** Ratings (1-10) were extracted directly from the filenames (e.g., `1234.7.txt` corresponds to a rating of 7).
- **Unsupervised Data:** 50,000 unlabelled reviews extracted from the `train/unsup` folder. This dataset serves as a sandbox for natural sentiment estimation and clustering via K-Means and GMM (pending final integration).

3 Text Cleaning Pipeline

To ensure the models focused entirely on linguistic features rather than HTML structure, we implemented the following regex-based sequence:

1. **HTML Unescaping:** Translates HTML entities back to their normal characters.
2. **Strip Line Breaks:** Target and remove line-break elements commonly injected during web scraping.
3. **Strip Remaining Tags:** Target and permanently strip all remaining HTML tags using regular expressions.
4. **Normalize Whitespace:** Collapse multiple consecutive space characters into a single space.

4 Exploratory Data Analysis (EDA)

Extensive EDA on all 100,000 reviews revealed critical meta-features that guided our architectural decisions:

- **N-Grams:** Unigrams alone fail to capture negation. Bigrams like “worst movie” or “highly recommend” are significantly stronger predictive indicators.
- **Punctuation Correlation:** Regression analysis indicated that extreme sentiment (1-star and 10-star reviews) utilizes significantly more punctuation (e.g., “!!!”) and capitalization than moderate reviews.
- **Review Length:** Reviews tend to be significantly longer the more polarizing the opinion is. Both 1-star and 10-star reviews averaged higher word counts than 5-star or 6-star reviews.

5 Deduplication

After aggressive cleaning, several reviews collapsed into identical strings. We performed strict deduplication on the `text_clean` column to prevent data leakage between the training and validation sets, thereby avoiding artificially inflated accuracy metrics.